Learner Workbook

Applied AI and Automation for Executives. One page for each session. Write in it, or type in Word.

How to use this workbook

Each page has the day's question, your steps for the paper exercise or practical, the fields to fill in, two questions to think about and one decision.
Sessions 1 to 11 each have a worksheet page after the session page, with the numbers for the paper exercise. The templates named on a page are in your Templates folder and the case company files are in Practice Files. The answers to class exercises are not in this workbook.
From Session 12 you work on your own workflow and organisation. Before then, use the case company for the exercises.

My workflow (fill in at Session 12)

The workflow or email task I will map	
Owner	
How often and how long it takes today	
What better means (a number)	
AI tool I already use	

Session 0, the introduction

AI and machine learning: the map before the route

What do I need to understand about AI before I decide anything about it?

Your steps for today

- Read the Course Handbook before Session 1.
- Write down which part of the 14-session map you most want to understand.
- Write any question from today that was not answered.

Fill in

The session I most want to understand	
One idea from today I could explain to a colleague	
A question from today that I still have	
My laptop and AI assistant account are ready (yes or no)	

Think about

Which field or industry use on today's slides is closest to my company?

Which role in the line from data to decision is missing in my company?

My decision

The one question I want answered first in the course.	

I keep: The Course Handbook, with today's ideas, a glossary and what to prepare. Read it before Session 1, which begins with how a machine sees.

Session 1

Seeing: pixels, features and neurons

How can a machine read a picture when all it has is numbers?

Your steps for today

- Score the five payments on the worksheet with weights 3, 2 and 1.
- Change the weight for a big amount to 4 and mark which payment changes.
- Write one place in your work where pictures or scans are read, and one check you would keep.

Fill in

Total score of each of the five payments	
Which payment changed when the weight became 4	
The weights I would choose, and why	
One place in my work where pictures or scans are read	
The check I would keep	

Think about

What would I ask a vendor who says their system reads our scans accurately?

Which of our images are unusual or poor, and would a system have seen those?

My decision

Use it now, use it with a checker, wait, or do not use it, with the reason.	

I keep: A one-page sheet: how a picture becomes numbers, the payment-risk neuron worked by hand, and the digit results.

Session 1: worksheet for the paper exercise

Score five payments

Five payments, one checklist. Tick each question 1 (yes) or 0 (no), multiply by its weight, add, and flag at 4 or more.

Part 1: weights 3, 2 and 1

	Amount	Payee	Time	Big x3	New x2	After hours x1	Total
P1	UGX 1,450,000	New	14:00				
P2	UGX 85,000	New	23:30				
P3	UGX 2,300,000	Known supplier	10:00				
P4	UGX 60,000	Known supplier	09:00				
P5	UGX 1,350,000	New	22:00				

Big amount means over UGX 1,200,000. After hours means outside 08:00 to 18:00. A person checks any payment that totals 4 or more.

Part 2: now give the big amount a weight of 4

	New total	Flagged?	Changed from Part 1?
P1			
P2			
P3			
P4			
P5			

Would you accept this change? Why?

Session 2

Learning from examples: types of machine learning, decision trees, nearest neighbours

How does a machine learn a rule from examples, and how much can I trust what it learned?

Your steps for today

- Fill in the distance table for the 10 past claims and the new claim.
- Vote with 3 neighbours, then with 1, and write both answers.
- Draw one straight line separating paid from queried and count the claims it gets wrong.
- Write one recurring decision in your work and how many past cases you have on record.

Fill in

Distance from the new claim to each past claim	
The vote with 3 neighbours, and with 1	
How many claims my line gets wrong	
A recurring decision in my work	
How many past cases I have on record	

Think about

Which would my auditor accept more easily: a rule or a precedent?

Do I have enough past cases, and are they labelled?

My decision

A tree, neighbours, neither or a person, with the number of past cases.	

I keep: A one-page sheet: the four types of learning, a tree read as rules, nearest neighbours, and the honest scores from today's run.

Session 2: worksheet for the paper exercise

Find the nearest claims

Claim N is new. Decide whether it will be paid or queried by looking at the most similar past claims. One square across is UGX 100,000. One square up is 5 days to file. Distance by city blocks: squares across plus squares up.

New claim N: 5 across, 4 up.

Claim	Across	Up	Outcome	Squares across from N	Squares up from N	Distance
A	2	1	paid			
B	3	2	paid			
C	4	1	paid			
D	4	5	queried			
E	5	6	queried			
F	7	3	paid			
G	6	4	paid			
H	8	6	queried			
I	7	2	paid			
J	9	3	queried			

Vote of the 3 nearest		**Vote of the nearest 1**	
Claims my straight line gets wrong		**Does the answer change?**	

Draw the claims on squared paper (across along the bottom, up along the side), mark paid and queried differently, circle the 3 nearest to N, and draw one straight line that best separates paid from queried.

Session 3

Measuring a model: accuracy, precision, recall, F1, ROC and AUC, thresholds, baselines, overfitting

How do I know whether a model is any good, and what should I ask the vendor who says it is?

Your steps for today

- The vendor question sheet: read all ten questions.
- The result table from the recorded run, with room for your notes.
- The twenty cases and the blank confusion table: fill it by hand.
- Your task: the costlier mistake and a first UGX figure for each.

Fill in

My task, in one line	
Which mistake costs more, a false alarm or a miss	
Rough UGX cost of a false alarm	
Rough UGX cost of a miss	
The first question I will put to a vendor	

Think about

Where have I seen a single accuracy figure used to sell something?

What would a simple baseline be for the task I chose?

My decision

The costlier mistake for my task, and the first number I will ask a vendor for.	

I keep: The vendor question sheet: ten questions about the numbers behind any AI claim. Keep it next to every proposal you read.

Session 3: worksheet for the paper exercise

Twenty cases and the four boxes

Twenty invoices from the test set of the recorded run. The model flags an invoice at a score of 0.20 or more. Late means paid late in fact.

No.	Invoice	Amount, UGX	Score	In fact	Flagged at 0.20? (Y/N)	Box (TP FP FN TN)
1	INV0318	470,000	0.33	On time		
2	INV0168	744,000	0.47	Late		
3	INV1940	822,000	0.10	On time		
4	INV1486	427,000	0.01	On time		
5	INV0216	463,000	0.07	On time		
6	INV1334	195,000	0.13	On time		
7	INV2427	774,000	0.17	Late		
8	INV1848	824,000	0.06	On time		
9	INV1345	218,000	0.02	On time		
10	INV0761	216,000	0.08	Late		
11	INV0002	524,000	0.08	On time		
12	INV2938	285,000	0.03	On time		
13	INV2286	464,000	0.11	On time		
14	INV1810	309,000	0.13	On time		
15	INV2788	889,000	0.09	Late		
16	INV0512	4,262,000	0.73	Late		
17	INV1030	224,000	0.03	On time		
18	INV1920	268,000	0.12	On time		
19	INV0655	697,000	0.53	Late		
20	INV1629	391,000	0.32	On time		

The four boxes and two measures

Caught (flagged and late)		**False alarms (flagged, on time)**	
Missed (not flagged, late)		**Rightly cleared (not flagged, on time)**	

The four numbers add up to 20. Precision = caught / flagged = _____. Recall = caught / really late = _____.

Move the line

Line	Flagged	Caught	Precision	Recall
0.50				
0.20				
0.15				

Which line would you choose, and why?

Session 4

Words as numbers: bag of words, TF-IDF, embeddings and distance

How can a computer find the right clause or policy when I describe it in plain words?

Your steps for today

- The distance cheat sheet: read it with the arrows in mind.
- The three toy arrows and the empty table. Fill in the twelve answers by hand.
- The recorded search results for the three kinds of text, with space for your notes.
- Three plain descriptions from your own work and your predictions.

Fill in

Three formal terms from my work	
A plain description of each, using none of its words	
Would a word search find it (yes, no, unsure)	
Would a search by meaning find it (yes, no, unsure)	
What a wrong first answer would cost me	

Think about

Where do I search a pile of text today, and what do I type?

Where must a search match exactly?

My decision

The pile of text where search by meaning would help most, and what a wrong first answer would cost.	

I keep: The distance cheat sheet: four ways to say how close two things are, each with its picture, its rule and when to use it.

Session 4: worksheet for the paper exercise

Four distances for three arrows

Three toy arrows, drawn from the starting point (0, 0). A is 3 across and 4 up. B is 3 across and 1 up. C is 6 across and 0 up. Mark them on squared paper.

Arrow	Across	Up	Length
A	3	4	5
B	3	1	3.16
C	6	0	6

Pair	Straight line	City blocks	Agreement score	Cosine
A with B				
A with C				
B with C				

Agreement score: multiply the across numbers, multiply the up numbers, add. Cosine: agreement score divided by (length of the first arrow x length of the second).

Which pair does each measure call closest? _____

Three searches from my own work

Formal term	Plain description (none of its words)	Word search finds it?	Meaning search finds it?

Session 5

Language models: tokens, attention and next-word probability

How does a language model turn words into an answer, and why can the same question give different answers?

Your steps for today

- Write a sentence from your work. Estimate its tokens, then check with the model if it is run live.
- Work the softmax by hand with scores 3, 2 and 0.
- Repeat after dividing the scores by 2 and compare.
- Write one task where variety helps and one where it hurts.

Fill in

My sentence and its token count	
Softmax percentages at temperature 1	
Softmax percentages at temperature 2	
A task where variety helps	
A task where it hurts	

Think about

Where in my work would a confident but wrong answer cost the most?

Which of my documents would a model need to see before it could answer questions about my company?

My decision

The task I choose, the category (draft, draft and check, needs a source, keep it human) and why.	

I keep: A one-page sheet: the next-word loop, the softmax recipe, what temperature does and five questions to ask about any language-model answer.

Session 5: worksheet for the paper exercise

A softmax and a temperature by hand

Turn scores into percentages. Raise 2.72 to the score, add the three answers, and divide each answer by the total. Scores: days 3, weeks 2, minutes 0.

Temperature 1 (scores as they are)

Piece	Score	2.72 to the score	Share of the total
days	3		
weeks	2		
minutes	0		

Temperature 2 (divide each score by 2 first)

Piece	Score / 2	2.72 to the score	Share of the total
days	1.5		
weeks	1		
minutes	0		

Which setting gives the safer, more repeatable answer, and which gives more variety?

Session 6

Foundation models: pretraining, post-training, model families, reasoning and multimodal models, benchmarks

What is a foundation model, how does it become an assistant, and what does a benchmark score tell me?

Your steps for today

- Explain pretraining and post-training in your own words, with the library and the coach.
- Answer the four model card questions for the GPT-2 card.
- Fill one row of the comparison grid for your own task.
- Write the three checks you will use and the ten cases you will test.

Fill in

My task in one line	
Models I can reach	
Three checks	
Ten test cases (list them)	
Model card: licence, size, limits	

Think about

Which claim from a vendor would I now ask to see tested on my own sample?

Where in my work would an invented promise cost the most?

My decision

The route I choose (hosted assistant, reasoning model, open weights or wait for evidence), my task and my three checks.	

I keep: A one-page model comparison grid for your own tasks, with the questions to ask of a model card and a benchmark claim.

Session 6: worksheet for the paper exercise

Read a model card

Read the printed model card (GPT-2, from its Hugging Face page) and answer four questions with a line from the card.

Question	Line from the card	Found? (Y/N)
What is the licence, and may you use it at work?		
How big is it, and what was it trained on?		
What does the maker say it should not be used for?		
What is missing that you would want to know?		

Session 7

Open weights, file formats, size and running cost

What does it mean to own, host or rent a model?

Your steps for today

- Multiply the numbers in a 7 billion model by 2 bytes and by half a byte.
- Name one task of yours and the data it touches.
- Choose rent, host or own, and write your reason.
- Write the licence question you would ask first.

Fill in

Memory sum at 16-bit (7 billion x 2 bytes)	
Memory sum at 4-bit (7 billion x 0.5 byte)	
My task and its data	
My route (rent, host or own) and my reason	
The licence question I would ask first	

Think about

Which of my tasks touch data that should not leave the company?

What would make me move from renting to hosting?

My decision

My route for my first model, and the one thing that would change my mind.	

I keep: A one-page rent, host or own decision sheet, with the memory sum and the licence questions to ask.

Session 7: worksheet for the paper exercise

The memory sum and the decision sheet

Memory for the numbers inside a model = number of numbers x bytes for each number. 1 billion numbers at 1 byte is 1 gigabyte (GB).

Model	Sum	GB
7 billion at 2 bytes (16-bit)	7 x 2 =	
7 billion at half a byte (4-bit)	7 x 0.5 =	
70 billion at 2 bytes (16-bit)	70 x 2 =	
70 billion at half a byte (4-bit)	70 x 0.5 =	

Decision sheet: one task of mine and its data, rent, host or own, my reason, and the licence question I would ask first.

Task and the data it touches	
Route (rent, host or own)	
Reason	
First licence question	

Session 8

Prompting and reasoning models

How do I get reliable work out of an assistant?

Your steps for today

- Take a prompt you use often and write it as a five-part brief.
- Write three cases with known answers to test it on.
- Run it, score it against the known answers and fill one row of the prompt log.
- Change one thing and run it again.

Fill in

My prompt, rewritten with role, context, example, format and checks	
Three cases with known answers	
Version, what changed, runs, how many right	
Who checks the figures before I use them	

Think about

Which of my prompts has a format I could ask for as a table or JSON?

Where could a calm, confident answer hide a slip of the size we saw today?

My decision

The prompt I will test first, and who checks it.	

I keep: A reusable prompt template and a prompt log, so that every prompt you keep has been tested.

Session 8: worksheet for the paper exercise

The five-part prompt

Weak prompt: "Reply to this customer complaint." Rewrite it using the five parts. Customer to answer: Mercy Atim, whose jerrycans of oil leaked for the third time (an NVF example).

Part	My sentence
Role and reader	
Context (what happened, which customer)	
One example, or 'none yet'	
Format and length	
Checks, and what to say when unsure	

Prompt log

Version	What changed	Runs	How many right	Who checked

Session 9

Giving a model your knowledge: retrieval-augmented generation, long context and fine-tuning

How do I give an AI model my company's knowledge, and when is each way worth it?

Your steps for today

- For each of the six NVF requests, write P, R or F and one line of reason.
- Write ten questions with known answers for one document in your job.
- Run the cheapest rung on your ten questions and count the right answers. Optional.

Fill in

The request from my job that needs my knowledge	
The rung I think it needs (prompt, retrieve, fine-tune)	
My ten questions and where each answer is found	
Right answers on the cheapest rung (out of 10)	
What would make me climb to the next rung	

Think about

Which of my documents change often, and who would keep an index up to date?

Do I hold examples I may use for fine-tuning, with personal data removed?

My decision

The rung I will try first, and the ten-question test I will write.	

I keep: A one-page decision tree: prompt first, then retrieve, then fine-tune, and the test that tells you when to climb.

Session 9: worksheet for the paper exercise

Choose the rung

Choose a rung for each request: P (prompt), R (retrieve) or F (fine-tune). Give one line of reason.

	Request	P, R or F	Reason
a	Summarise one supplier email.		
b	Answer staff questions from a handbook that changes each quarter.		
c	Every customer reply ends with the house sign-off.		
d	Check a price against one pasted quote.		
e	Answer questions about 400 contracts.		
f	Draft replies in the Sales Manager voice from 200 past emails.		

The ten-question test I would run first on the cheapest rung: write it on the Session 10 test-set page or on paper.

Session 10

When models fail, and how to check them

How do I find out when an AI answer is wrong, before it costs me?

Your steps for today

- Score the practice answer sheet against the key: right, wrong, missed, invented.
- Write your own ten-question test set for one document in your job.
- Run it on one assistant, twice, and score it. Optional.

Fill in

The document I chose	
My ten questions, known answers and sources	
Score on run 1 and run 2 (right, missed, invented)	
The cost of one wrong answer, in UGX or in time	
Who signs off the answers	

Think about

Which of the five cases could happen in my own work this month?

Which question in my set would catch an invented answer?

My decision

The level of checking for one AI use in my job, and the first check I will make.	

I keep: A five-step check routine and a test-set template, each on one page.

Session 10: worksheet for the paper exercise

Score an answer sheet and write a test set

Mark each answer on the practice sheet against the key: right, wrong, missed or invented. The key is handed out in class.

Q	Question	Sheet says	Key says	Mark
Q01	How many working days of paid annual leave does a full-time staff member earn in each leave year?	21 working days (clause 2.1).		
Q02	How long before the first day of leave must a leave request reach the line manager?	At least 14 days before the first day of leave (clause 2.2).		
Q03	Who must approve an expense claim of UGX 1,500,000?	The Managing Director (clause 3.2).		
Q04	Within how many days of spending must receipts reach the Finance office?	Within 14 days of the spending (clause 3.1).		
Q05	How is approved overtime paid on a Sunday?	1.5 times the normal hourly rate (clause 1.3).		
Q06	May a staff member paste a customer price list into a public AI tool?	No, unless the Managing Director has approved the tool in writing (clause 4.4).		
Q07	How soon must a lost company laptop be reported, and to whom?	Within 4 hours of discovery, to the Operations Manager (clause 5.3).		
Q08	What is the most the company will pay for one night of accommodation on company travel?	UGX 180,000 a night (clause 3.3).		
Q09	How many days of paid sick leave does a staff member get each year?	10 days (clause 2.4).		
Q10	What rate per kilometre does the company pay when a staff member uses a private car for work?	Not stated in the handbook.		

Right ___ Wrong ___ Missed ___ Invented ___. Which error would cost most, and why?

_____

My ten-question test set

Id	Question	Known answer	Source	Run 1 / Run 2	Cost of a wrong answer
Q1					
Q2					
Q3					
Q4					
Q5					
Q6					

Id	Question	Known answer	Source	Run 1 / Run 2	Cost of a wrong answer
Q7					
Q8					
Q9					
Q10					

Two of the ten questions should have no answer in the document. The right answer to those is "not stated".

Session 11

Agentic AI: tools, loops, MCP, autonomy and guardrails

What changes when an assistant can act as well as answer?

Your steps for today

- Fill the four rows of a Tool Boundary Sheet for the mailbox assistant.
- Name the row that stops M02, M03, M07 and M08.
- Trace plan, act and observe for M01, M03 and M08.
- Count the approvals and your minutes.

Fill in

The task I would give an agent	
The level of autonomy I would allow	
Badges it needs (read, write, send, spend)	
Who approves, and the off switch	
Approval minutes a week (my estimate)	

Think about

Which tool that I already use could act on my behalf today?

Who would notice first if an agent in my company went wrong?

My decision

The level of autonomy I allow for my task, and the one badge I will never give.	

I keep: Your Tool Boundary Sheet: what an agent may read, write, send and spend, who approves, the log and the off switch.

Session 11: worksheet for the paper exercise

The boundary sheet and the loop

Fill the four rows for the mailbox assistant, then name the row that stops each risky mail. The printed mails M01, M03 and M08 come from the demonstration mailbox (all invented).

Row	It may	A person approves	It must never
Read			
Write			
Send			
Spend			

Mail	Plan	Act (one tool)	Observe, and where the loop must stop
M01			
M03			
M08			

Approvals a week x minutes each = _____ minutes a week. Owner: _____ Off switch: _____

Session 12

Workflow mapping (practical)

Which of my tasks should be a rule, an AI step or an agent?

Your steps for today

- Map one weekly task in lanes: trigger, steps, handoffs, check.
- Tag each step: rule, AI step, agent or person, with a reason and a check.
- Sketch one assistant with its four boundary rows, if one fits.
- Fill the Inputs sheet of the Business Case Workbook and mark E or M.

Fill in

My workflow and its trigger	
The handoff that causes the most waiting	
The step I would hand to AI first, and how I check it	
Hours saved a month (plan)	
The four numbers I will record for four weeks	

Think about

Which step on my map is a rule I could set up this week without AI?

Who else in my organisation runs a workflow like mine?

My decision

What I build or test first, and the number that will show it worked.	

I keep: Your mapped workflow, the choice for each step, your assistant sketch and the hours-saved estimate in the Business Case Workbook.

Session 13

Governance and buying (practical)

What rules must my company set for AI, and how do I question what I am sold?

Your steps for today

- Edit the Risk Register for your own work: a score, an owner and a control for your top five.
- Draft sections 3, 4, 5, 7, 8 and 10 of the AI Use Policy Template.
- Answer three groups of the Vendor Question List for one tool, and write five questions with red-flag answers.
- Fill in the lawyer page and name the policy owner, a deputy and the review date.

Fill in

My work and the tool I chose	
My top five risks, their scores and levels	
Owner and control for each	
Approved tools in my policy, each with an owner	
What may go in at each data class	
What always needs a named reviewer	
Who hears about an incident, and how fast	
My three biggest gaps about the tool	
My five vendor questions and the red-flag answer for each	
Which of my AI uses involve personal data	
Policy owner, deputy and review date	

Think about

Which risk surprised me when I scored it?

Who in my company already uses AI through a personal account?

Which vendor claim would I have believed before today?

My decision

Decline, ask in writing or run a pilot: my route for the next proposal, the reason and the first step.	

I keep: Your Risk Register, your AI Use Policy draft, your Vendor Question List with five questions, and your page of questions for the lawyer.

Session 14

Data readiness assessment, roadmap and 90-day plan (practical)

How ready is my organisation's data for AI, what will I fix first, and who will own it?

Your steps for today

- Build the Information Inventory: at least six sources, each owned and labelled.
- Score the six areas Green, Amber or Red with evidence, and read the total out of 12.
- Turn your top three gaps into at least six roadmap actions with owners, dates, hours and measures.
- Put the day 30, 60 and 90 check-ins in your diary and list three risks to the plan.

Fill in

My organisation and my plan start date	
Sources in my inventory (number) and how many are Confidential or Personal	
My six ratings and my total out of 12	
My top three gaps, with evidence and a first fix	
My roadmap actions, each with owner, due date, hours and measure	
Decisions the company must make, who decides and by when	
Three risks to the plan and what I will do	
My check-in dates: day 30, day 60, day 90	
How many gaps I will close by day 90	
The action I will take within 24 hours	

Think about

Which area surprised me when I scored it?

Which of the fourteen sessions changed how I think most, and why?

What do I now understand about AI before I decide anything about it?

Who must hear about this plan first, and what do I want from them?

My decision

How many gaps my roadmap closes by day 90, and my first action within 24 hours.	

I keep: Your inventory and scorecard, your roadmap with its risks and decisions, and the Follow-up Kit: Check-in Form, Results Tracker and Sponsor Briefing.

Words used in this course

Word	What it means	First used
Model	The thing that has learned patterns and makes the predictions.	Session 0
Parameter	One of the many settings inside a model that training adjusts. Also called a weight.	Session 0
Training	Showing the model many examples and adjusting its settings until its errors shrink.	Session 0
Inference	Using a trained model to make a prediction or write an answer.	Session 0
Overfitting	When a model memorises its examples and fails on new ones.	Session 0
Pixel	One square of a picture, stored as a number (grey) or three numbers (colour).	Session 1
Feature	A measurement taken from the input, such as the amount of ink in the middle.	Session 1
Neuron	A weighted checklist: it multiplies, adds and compares to give one answer.	Session 1
Weight	How much one question counts in a neuron. Training adjusts the weights.	Session 1
Layer	A row of neurons that look at the same inputs. A network stacks layers.	Session 1
Label	The right answer attached to an example, such as order or payment.	Session 2
Classification	Choosing a category for each case. Predicting a number is called regression.	Session 2
Decision tree	A set of yes-or-no questions that leads from a case to an answer.	Session 2
Nearest neighbours	Answer a new case by looking at the most similar past cases.	Session 2
Cross-validation	Taking turns to hide part of the data so the score is honest.	Session 2
Baseline	A simple way of doing the task that a model must beat, such as always saying no.	Session 3
Threshold	The score above which the model raises an alert. You choose it.	Session 3
Precision	Of the cases the model flagged, the share that really were the thing.	Session 3
Recall	Of the real cases, the share the model flagged.	Session 3
AUC	One number for how well a model ranks real cases above the rest, at every threshold.	Session 3
Bag of words	A message turned into a tally of how often each word appears. The order is lost.	Session 4
TF-IDF	A tally in which rare words count for more than common ones.	Session 4
Embedding	A list of numbers that places a piece of text on a map of meaning.	Session 4
Vector	Any list of numbers. Drawn as an arrow from the starting point.	Session 4
Cosine similarity	How close the direction of two arrows is. 1 means the same way.	Session 4
Token	A piece of text a model reads and writes: a short word, part of a long word, a number or a mark.	Session 5
Softmax	A recipe that turns scores into percentages that add up to 100.	Session 5
Temperature	A dial that sharpens or flattens the percentages before a piece is picked.	Session 5
Attention	The step where each word looks back at earlier words and weighs which ones matter.	Session 5
Context window	The most text, counted in tokens, that a model can look at in one go.	Session 5
Foundation model	A large model trained on a vast amount of text, images or both, that many products are built on.	Session 6

Word	What it means	First used
Pretraining	The first stage: reading huge amounts of data to learn to guess the next piece.	Session 6
Post-training	The coaching stage that turns a base model into an assistant that follows instructions.	Session 6
Benchmark	A fixed set of questions with a score, used to compare models.	Session 6
Model card	The page a maker publishes about a model: licence, size, uses and limits.	Session 6
Open weights	A model whose trained numbers you may download and run, under a licence that can still limit use.	Session 7
Weights	The numbers inside a model, set during training. Also called parameters.	Session 7
Quantisation	Rounding each weight to fewer allowed values, so each one needs fewer bits and the file shrinks.	Session 7
Container	The file format that packs the numbers: Safetensors, GGUF or ONNX.	Session 7
Hosting	Running a model on a server you rent, so you choose where your data goes.	Session 7
Prompt	Anything you type to an assistant.	Session 8
Brief	A prompt written once, in parts, and reused. Like a task handed to a new colleague.	Session 8
Example prompt	A finished example inside the prompt that shows the format you want. Also called a few-shot prompt.	Session 8
Structured output	An answer in a fixed shape, such as a table or JSON, so that it can be checked.	Session 8
Reasoning model	A model that works through a problem before it replies. Some assistants let you switch this on.	Session 8
Retrieval	A search step that fetches the right passages before the model answers. Short for retrieval-augmented generation, or RAG.	Session 9
Chunk	One piece of a document, cut so that it can be searched.	Session 9
Fine-tuning	Extra training that changes the habits of a model. LoRA is a light way to do it.	Session 9
Failure	An answer that is wrong, invented, unsupported or unsafe to act on.	Session 10
Test set	Questions with known answers that you run to score a tool.	Session 10
Key	The known answers, with the source of each. Also called ground truth.	Session 10
Prompt injection	Text in the input that tries to steer the model. It is data and never an order.	Session 10
Agent	An AI model given a goal and tools, which chooses its own next step in a loop.	Session 11
Tool	One thing an agent can do outside the chat, such as read a mail or save a draft.	Session 11
Least privilege	Giving an agent the smallest set of permissions that still lets it do the job.	Session 11
Planted instruction	Text inside a mail, page or file that tries to give orders to the AI reading it. Also called prompt injection.	Session 11
MCP	Model Context Protocol: an open standard for plugging tools and data into AI applications.	Session 11
Workflow	A task that repeats, from the event that starts it to the result someone checks.	Session 12
Trigger	The event that starts a workflow, such as a message arriving.	Session 12
Handoff	The moment work passes from one person or system to another. Errors and delays gather here.	Session 12
Governance	The rules, owners and checks that keep AI use safe and accountable.	Session 13

Word	What it means	First used
Risk register	A table of what could go wrong, how likely and how bad, who owns it and what controls it.	Session 13
Control	A real action that lowers a risk, such as a named person who checks the figures.	Session 13
Red flag	A claim, a term or a silence in a proposal that is a reason to question the vendor.	Session 13
Pilot	A small, time-limited trial on your own data, with a baseline and a rule for when to stop.	Session 13
Data readiness	How far your information is readable, findable, complete and allowed to be used with AI.	Session 14
Inventory	A written list of what information you hold, who owns it, where it lives and who may see it.	Session 14
Metadata	Facts about a file, such as date, owner and version, that help you find and trust it.	Session 14
Roadmap	A plan of actions, each with an owner, dates, hours and a way to check it worked.	Session 14
Check-in	A short review at day 30, 60 and 90 of your plan, with a test and a decision.	Session 14